

All words that donot start with 'a' over $\Sigma = \{a, b\}$

$f = \{b, ba, baa, bba, baab, bbaa, bbbba, baabb, \dots\}$

regular expression = ~~$b(a+b)^*$~~

All words that start with double letter over $\Sigma = \{a, b\}$

$f = \{aa, bb, aab, bba, aabbb, bbaaa, aaabb, aaa, bbb, \dots\}$

regular expression = $(aa+bb)(a+b)^*$
OR

$aa(a+b)^* + bb(a+b)^*$

15-10-2024

$a(a+b)^*$

* $\rightarrow b^* = \lambda, b, bb, bbb, \dots$

$\rightarrow (ab)^* = \lambda, ab, abab, ababab, \dots$

$\rightarrow (a+b)^* = \lambda, a, b, ab, ba, aa, bb, \dots$

$\rightarrow (a^*+b) = \underline{b}, \underline{\lambda}, a, aa, aaa, \dots$

multiple choice $\rightarrow$ one choice $\rightarrow$ only b.

$(a+b)^2 = (a \cup b)^2$
 ↳ choice ↳

→ $L = \{ a, ab, abb, abbb, abbbb, \dots \}$
 ↳ regular expression = $(ab)^+$

$(ab)^+ \neq a^+b^+$
 ↳ $(ab)^+ = \{ \lambda, ab, abab, ababab, \dots \}$
 ↳ $a^+b^+ = \{ \lambda, a, b, abb, aabb, aaabbb, \dots \}$
 ↳ different words are generated from these that is why these are not equal.

$L = \{ x, xx, xxx, xxxx, xxxxx, \dots \}$

Regular expression = xx^+

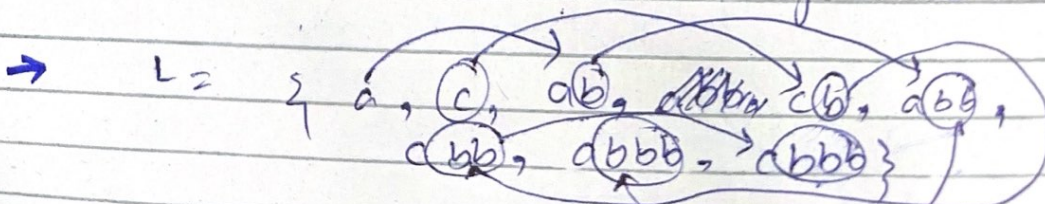
xx^+
 = x

OR

R.E = x^+

$xx^+ = x^+$

same words are generated



→ a K baad 1 b phir 2 b phir
 3 b. phir

Regular expression = $(a+c)^+b$.

or
 $(ab^+ + cb^+)$

consider a finite language L
that contains all the strings
of a 's and b 's of length
three exactly.

$L = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$

Regular Expression = $(a+b)(a+b)(a+b)$

we have choice
to ~~stage~~ choose
 a or b from
each.

$\left[\begin{array}{l} a a a \\ a a b \\ a b a \\ a b b \\ b a a \\ b a b \\ b b a \\ b b b \end{array} \right]$

Regular Expression
by Recursive Definition.

r_1

$r_1 r_2$

$r_1 + r_2$

$(r_1 + r_2)^*$

i.e. $\forall$

→ All will remain
regular expression.

$$a(b+a)^* = \{ a, aa, ab, aab, aba, \dots \}$$

$$(a+b)^* = (b+a)^*$$

$$\{ bba, bbb \} = bb(a+b)^*$$

regular expression.

All words that end with ba .

$$\rightarrow (a+b)^* ba$$

All words that contain at least one a .

$$\rightarrow (a+b)^* a (a+b)^*$$

All words that contain at least two a .

$$\rightarrow (a+b)^* aa (a+b)^*$$

Write RE:

$$\Sigma = \{ a, b \}$$

① All words ending with b .

$$(a+b)^* b$$

$$\{ b, ab, abb, abbb, \dots \}$$

② All words start with a .

$$a(a+b)^*$$

③ All words that start with double letter.

$$aa(a+b)^* + bb(a+b)^*$$

OR

$$(aa+bb)(a+b)^*$$

(+) $\rightarrow$ 1 k baar chley ga he chley ga.

(4) All words that contain one double letter.

$L = \{ aa, bb, abb, baa, abba, baab, abababb, babaaa, \dots \}$

$$R.E = (a+b)^* (aa+bb) (a+b)^*$$

(5) All words that end and start with double letters.

$$R.E = (aa+bb) (a+b)^* (aa+bb) (aa+bb+aaa+bbb)$$

(6) All words of length ≥ 3

$$(a+b)(a+b)(a+b)$$

$$R.E = ((a+b)(a+b)(a+b))^+ (a+b)^*$$

$L = \{ aaa, bbb, abb, \dots, abbba, aaaa \}$

OR

$$R.E = (a+b)(a+b)(a+b)^+$$

(7) All words of length $L=3$

$L = \{ a, b, ab, ba, aa, bb, aba, abb, aab, bab, bba, bbb \}$

$$R.E = ((a+aa+ab+ba+bb+aaa+bbb) + (a+b)(a+b)(a+b))^* \rightarrow \text{loop infinite}$$

(8) All words that contain exactly one a or exactly one b.

$$R.E = (b^*ab^* + a^*ba^*)$$

↓

$\lambda a \lambda$

$\lambda b \lambda$

$\geq a$

$b.$

⑨ All words that don't end at b.

$$(a+b)^* a$$